

Module-1

Lecture-3: Stress - Strain Relationship

MECHANICS OF SOLIDS

- Stress strain Relation in 1-D system
- Stress strain Relation in 2-D system
 - Longitudinal strain
 - Lateral strain
 - Poisson's ratio
 - Stress strain Relation
- Stress strain Relation in 3-D system
- Relation b/w elastic constants

Stress strain Relation in 1-D system

- Relationship between stress and strain for a unidirectional stress is given by **Hooke's Law**

$$E = \frac{\sigma}{\epsilon}$$

$$G = \frac{\tau}{\phi}$$

Stress strain Relation in 2-D system

Longitudinal strain

- Strain along line of action of applied force
- Longitudinal strain = Increase in length/Original length
- $\epsilon_{longitudinal} = \frac{\delta l}{l} = \frac{l-l_0}{l_0}$

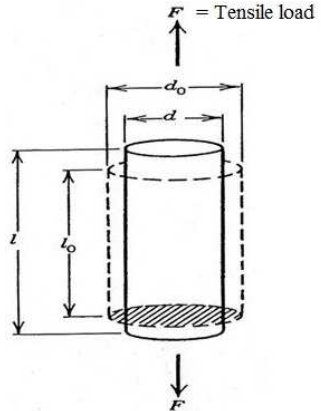
Lateral strain

- Strain at right angle to the line of action of applied force
- Lateral strain = Decrease in dia/Original dia
- $\epsilon_{lateral} = \frac{\delta d}{d} = \frac{d-d_0}{d_0}$

Poisson's ratio(μ)

$$\mu = \frac{\epsilon_{lateral}}{\epsilon_{longitudinal}}$$

$$\epsilon_{lateral} = -\mu * \epsilon_{longitudinal}$$



2-D stress strain relation

Strain produced by σ_x

- $\epsilon_{x1} = \frac{\sigma_x}{E}$
- $\epsilon_{y1} = -\mu \frac{\sigma_x}{E}$

Strain produced by σ_y

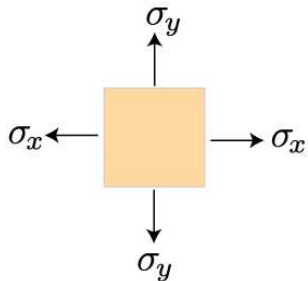
- $\epsilon_{y2} = \frac{\sigma_y}{E}$
- $\epsilon_{x2} = -\mu \frac{\sigma_y}{E}$

Total strain along x & y axis

$$\epsilon_x = \epsilon_{x1} + \epsilon_{x2} \quad \epsilon_y = \epsilon_{y1} + \epsilon_{y2}$$

$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_x}{E}$$



3-D stress strain relation

Strain produced by σ_x

- $\epsilon_{x1} = \frac{\sigma_x}{E}$ $\epsilon_{y1} = \epsilon_{z1} = -\mu \frac{\sigma_x}{E}$

Strain produced by σ_y

- $\epsilon_{y2} = \frac{\sigma_y}{E}$ $\epsilon_{x2} = \epsilon_{z2} = -\mu \frac{\sigma_y}{E}$

Strain produced by σ_z

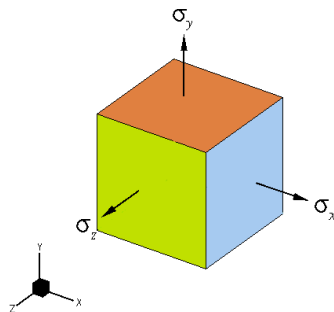
- $\epsilon_{z3} = \frac{\sigma_z}{E}$ $\epsilon_{x3} = \epsilon_{y3} = -\mu \frac{\sigma_z}{E}$

Total strain along x,y & z axis

$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_z}{E}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E}$$



Volumetric Strain

Ratio of Change in volume to original volume of a deforming body

$$\epsilon_v = \frac{\delta V}{V}$$

where: δV = Change in volume
 V = Original volume

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

where: ϵ_x = Strain along X axis
 ϵ_y = Strain along Y axis
 ϵ_z = Strain along Z axis

Relation b/w elastic constants

Young's modulus - Bulk modulus

$$E = 3K(1 - 2\mu)$$

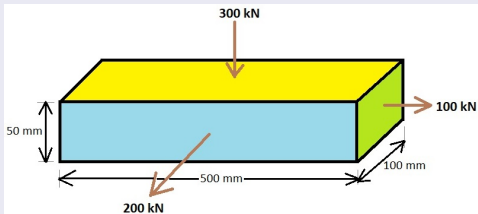
Young's modulus - Rigidity modulus

$$E = 2G(1 + \mu)$$

- ① A rod 150cm long and 2cm dia is subjected to an axial pull of 20kN. If elastic modulus of material is $0.2\text{MN}/\text{mm}^2$, determine the stress, strain and elongation of the rod
- ② Find the minimum diameter of a steel wire, which is used to raise a load of 4000N if the stress in the rod is not to exceed $95\text{MN}/\text{m}^2$?
- ③ Find the Young's modulus of a brass rod of diameter 25mm and length 250mm, which extends 0.3mm on a tensile load of 50kN ?
- ④ The safe stress for a hollow steel column which carries an axial load of 2.1MN is $125\text{MN}/\text{m}^2$. If the external dia of column is 30cm, find the internal dia ?
- ⑤ Determine the change in length, breadth and thickness of a steel bar, which is 4m long, 30mm wide and 20mm thick and is subjected to an axial pull of 30kN along the direction of length. Take $G=77\text{kN}/\text{mm}^2$ and $\mu=0.3$?

Problem-1

A rectangular bar is subjected to forces as in figure. What is the change in volume of the bar? Take $E = 200 \text{ GPa}$, Poisson's ratio $= 0.25$



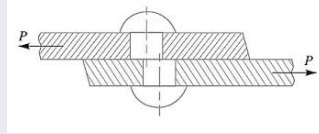
Problem-2

A steel cube of 50 mm side is subjected to a force of 6 kN (Tension), 8 kN (Compression) and 4 kN (Compression) along three mutually perpendicular axis. Determine the change in volume of the block. Take E as 200 GPa and m as $10/3$

Shear Stress

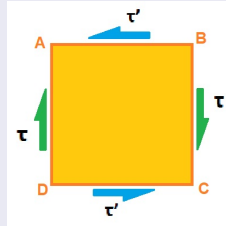
Shear Stress

When a section is subjected to two equal and opposite forces, acting tangentially across a resisting section, as a result of which the body tends to shear off across the section, the stress induced is called the **Shear stress**. The corresponding strain is called **Shear strain**



Principle of Shear Stress

A shear stress across a plane, is always accompanied by a balancing shear stress across the plane and normal to it



*Thank
you!*